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VALUING MULTIPLE TRAJECTORIES OF KNOWLEDGE: A CRITICAL REVIEW AND AGENDA FOR KNOWLEDGE MANAGEMENT RESEARCH

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Over the past three decades, scholars have increasingly come to view knowledge as one of the most important resources necessary for successful organization in the contemporary socioeconomic landscape. In our vigor to understand how organizations may harness the diverse knowledge available to them, however, we have produced a disparity in our theories of knowledge management (KM) processes. By reviewing 20 years of influential KM literature, we uncover a bias toward explaining knowledge integration over research exploring processes of knowledge differentiation. Through our review, we explain why such a pattern has emerged and build an argument for why understanding differentiation is an increasingly important charge for management and organizational scholars. We then advance three strategic directions for future KM scholarship, based on the notion that recognizing multiple knowledge trajectories can aid in addressing several significant lines of theorizing in management and organization studies.

The tension between integrated and differentiated knowledge has been central to organizational theory for the better part of a century. One of the fundamental advantages of a classic bureaucratic organizational structure is its capacity to support subunits whose differentiated knowledge, knowledge unique to their specialized role and position, enables the efficient distribution of organizational tasks (Galbraith, 1973; Weber, 1978) and exploration of opportunities to expand organizational knowledge (March & Simon, 1958). Alternately, integrated knowledge, that which is common among organizational subunits, can also facilitate important organizational processes such as coordinating action (Cyert & March, 1963; Padgett, 1980a, 1980b), supporting organizational learning and adaptation (Nelson & Winter, 1982), and spurring innovation (Hayek, 1945; Nonaka, 1994; Nonaka & Takeuchi, 1995). Balancing the pressure governing movement toward these two knowledge states has proven to be one of the major challenges of contemporary organizing (March, 1991), and the need to manage this balance has become increasingly necessary as we move deeper into the knowledge economy (Bell, 1973; Drucker, 1994; Stark, 2009). Recognizing the centrality of knowledge in organizational practices, scholars developed a focus on knowledge management (KM) as an area of research primarily devoted to understanding the processes by which knowledge shifts from differentiated to integrated states to produce operational benefits (Davenport & Prusak, 1998; Spender & Grant, 1996; Szulanski, 1996).

It has become commonplace for organizations to engage in structural, technological, and relational interventions explicitly designed to harness knowledge to produce distinct value. These efforts are spurred by the belief that KM can allow organizations to leverage differentiated knowledge held by their employees, to integrate knowledge that may be ignored or which exists outside the organization, to generate new knowledge, or to move and apply knowledge in ways that advance organizational

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goals. Given these operational aims, scholars frequently view KM as a strategic endeavor: a plan of action that organizations pursue based on the conviction that managing the flow of knowledge can create value (Davenport & Prusak, 1998; Wiig, 1997).

Yet, the processes governing the production, flow, and maintenance of organizational knowledge have proven to be multiplex. Knowledge can reside in multiple locations, including individuals, bodies, routines, technologies, and practices (Blackler, 1995; Collins, 1993; Nelson & Winter, 1982), which creates challenges when attempting to make sense of how knowledge operates in a specific organizational context. Furthermore, knowledge is managed through a variety of social and technical processes. Individuals can share and produce knowledge through interaction (Kuhn & Jackson, 2008; Tsoukas, 2009) and embody knowledge through virtual tools (Vaast & Walsham, 2005; Wasko & Faraj, 2005). Organizations can codify knowledge in organizational routines and structures (Nelson & Winter, 1982) and develop new knowledge through specialization and strategic alliances (Lavie & Rosenkopf, 2006). On top of these complexities, organizational knowledge can also be deeply ambiguous and ephemeral, such that locating and developing it can become a managerial challenge (Baumard, 1996). Scholarly interests in KM include both revealing the mechanisms governing organizational knowledge, reflecting critically on how those mechanisms interact, and examining how the character of organizational knowledge makes these interests both complex and evolving.

This article reviews 20 years of influential KM research with three fundamental goals. Our first goal is to inductively develop a meta-theoretical framework for understanding how scholarship has examined the processes governing the trajectory of knowledge between integrated and differentiated states. Our second goal is to uncover any emergent patterns in how these processes have been emphasized in our literature. Finally, by taking stock of how we have approached KM, we seek to identify opportunities to extend KM research and to build an agenda for ongoing KM scholarship.

Our initial analysis produces a framework identifying four trajectories of knowledge in organizations, each governing a distinct flow of knowledge between integrated and differentiated states. We address our second goal by returning to KM literature to assess the prevalence of each trajectory in management and organization studies (MOS) research. Our results show an imbalance in our theories of KM processes. We uncover a bias toward research explaining the mechanisms by which organizational subunits integrate diverse forms of knowledge over research exploring the mechanisms of knowledge differentiation. Finally, we consider the implications of such a bias for understanding organizational processes and performance. We offer three directions for future research that recognize the importance of, and offer a path for, studying multiple trajectories of knowledge in KM.

A BRIEF INTRODUCTION TO KNOWLEDGE MANAGEMENT

It has become trite to assert that knowledge is a fundamental source of benefit for contemporary organizations. The post-industrial economy described by Bell (1973) is now an accepted reality exemplified by the ubiquity of, and dependence on, information and communication technologies as a source and outcome of knowledge production (Castells, 2000). More than two decades ago, building on an emerging interest in the ways organizations could strategically manage knowledge, scholars such as Grant (1996a, 1996b), Spender (1994, 1996) and Nonaka (Nonaka, 1994; Nonaka & Takeuchi, 1995) advanced knowledge-based views of firms. This work argued that the utilization of knowledge by individuals and organizations serves as the key driver of sustained competitive advantage (Grant, 1996b; Kogut & Zander, 1992; Okhuysen & Eisenhardt, 2002; Spender, 1996). The foregrounding of knowledge drew attention to *knowledge-intensive* organizations, which rely more on intangible assets, services, and human capital than the more traditional capital infrastructure, raw materials, and formal procedures (Starbuck, 1992). A knowledge-based view of the firm consists not of a single stance, but comprises a constellation of views regarding the function of knowledge in organizational contexts and the associated efforts to manage this knowledge. For instance, one knowledge-based view was an extension of the resource-based view of the firm (Barney, 1991; Penrose, 1959) and foregrounded knowledge as the essential resource providing firms value. An alternate view focused on the ways in which firms were constituted by the actions of individuals creating and applying situated knowledge (Nonaka & Takeuchi, 1995; Spender, 1996). Despite varying views of how knowledge is enacted in organizations, these scholars all positioned the management of knowledge as a critical challenge, and opportunity, for organizations.

In recognizing the strategic value of knowledge, this early work sparked problem-oriented inquiry into the practices that would cause an organization to be successful in managing knowledge. This line of inquiry is now referred to as KM. Managing knowledge became viewed not only as something that organizations do organically, but a distinct domain of management subject to investments of technology, human capital, and financial resources (Alavi & Leidner, 2001; Davenport & Prusak, 1998; Empson, 2001; Grover & Davenport, 2001). KM developed as an umbrella term for a variety of processes, practices, and artifacts present in organizational contexts (Alvesson & Kärreman, 2001; Teece, 1998) designed to derive value through the application and utilization of knowledge. The questions for analysts then turned to the conceptions of knowledge characterizing these activities, as well as the dynamics of the processes by which KM is enacted. We take these up in the next section.

Plurality in Definitions of Knowledge

Although scholars have largely accepted the importance of KM, there is tremendous diversity in definitions of knowledge (for discussion of typologies of knowledge, see Alavi & Leidner, 2001; Blackler, 1995). For some, knowledge is equated with information, with the primary distinction being that knowledge can exist solely in the minds of individuals (Alavi & Leidner, 2001; Gorman, 2002). Nonaka (1996) advances the epistemological stance that knowledge is "justified true belief." Although this view can position knowledge as a commodity that can be represented and shared with others (Grant, 1996b), it also enables a dynamic view of knowledge in that individuals vary in their personal beliefs and acceptable justifications. Alternately, knowledge has been defined in terms of the ability to take action in a particular context (Mason & Apte, 2005; Orlikowski, 2002). Broadly, differences in views of knowledge can be characterized by their positions on three interrelated attributes: (a) whether knowledge is explicit, (b) where knowledge resides, and (c) how knowledge is enacted.

Perhaps, the most prominent distinction regarding the character of knowledge is between explicit knowledge, which can be codified and represented in some material form, and tacit knowledge, which is intangible and tied to a specific context. Explicit knowledge can be externalized from individuals through the use of symbols, objects, and language in a manner allowing transmission or display to others. Alternately, tacit knowledge refers to that which people know and apply in action, yet find difficult to articulate and share (Polanyi, 1966). Tacit knowledge is developed through socialization, experience, and situated practice in a specific context—and individuals may not even be aware they are using this knowledge. The recognition that some forms of knowledge can be codified and articulated, yet others cannot, also means that knowledge can be viewed as residing both within and beyond individuals.

Organizational knowledge can be viewed as residing in a variety of contexts varying in their abstraction, visibility, and accessibility. For instance, Blackler (1995) notes that knowledge can be located in individuals' minds (embrained knowledge), bodies (embodied), organizational routines (embedded), dialogue (encultured), and symbols (encoded). Nelson and Winter (1982) argue knowledge becomes located in the rules, routines, and procedures of organizations over time. At a more fundamental level, scholars differ in views of whether knowledge should always be viewed as residing with individuals (Grant, 1996b; Simon, 1997), or whether knowledge can belong to the organization (Hargadon & Fanelli, 2002; Nelson & Winter, 1982). Alternately, others view knowledge as capable of moving from individuals to the organization such that it can be accessed or shared by others. This perspective captures an important corollary of the idea that knowledge can reside in different locations, which is that various forms of knowledge differ in the ease with which they can be accessed and moved (Grant, 1996b).

In addition, perspectives diverge concerning whether knowledge is viewed as a static or dynamic resource or, put differently, should knowledge be viewed as *formed* in some manner or always in a state of *becoming*? Belying this distinction, scholars viewing knowledge as becoming, or enacted in action, often advocate an analytical focus on *knowing* as opposed to knowledge (Blackler, 1995; Kuhn & Jackson, 2008). The term knowledge tends to connote an object orientation: knowledge is something that does or can exist and is something to which action can be directed. Knowledge, understood as a resource to be possessed by organizations and individuals, can be captured, created, transferred, stored, and retrieved. Alternately, when the concern is *knowing,* the focus is on the actions individuals and groups take to meet their situated needs. This perspective views organizational knowledge as inextricably linked to the practices of individual workers as they seek to address emergent problems (Amin & Roberts, 2008). As we shall describe in greater detail, when the analytical concern foregrounds knowing, knowledge is portrayed as always open to change and contestation.

This plurality of definitions produces both value and challenges for KM scholars. Although each perspective on knowledge is connected to distinct conceptualizations of how it can be deployed to provide organizational benefits, this polysemy also produces analytic divides that make it difficult for scholars to generate insights across perspectives. Accordingly, in this article, we remain agnostic with respect to any specific definition of knowledge. To do otherwise would lead a review like ours to disregard swaths of the KM literature opting for an alternative view than any singular one chosen and, perhaps more importantly, fail to recognize the multiple objectives served by KM activities. The goal of our review is not to assert a particular view of how knowledge should be conceptualized, but to observe how extant definitions of knowledge have shaped scholarly treatments of KM and the related implications of these patterns. Our approach offers the potential to trace emergent patterns in the literature and to develop a meta-theoretical framework acknowledging the multifaceted character of knowledge and knowing in organizational practice.

REVIEWING TWO DECADES OF KNOWLEDGE MANAGEMENT RESEARCH

We performed a structured literature review evaluating how knowledge management has been represented in influential scholarly work in MOS over the past twenty years. To do this, we sampled the ten most cited articles on KM for each year from 1996 to 2015. We chose 1996 as a meaningful start point because this year marked the publication of the now canonical special issue of *Strategic Management Journal* on Knowledge and the Firm coedited by Grant and Spender (Spender & Grant, 1996). This special issue marked one of the first collections of essays specifically aimed at *managing* knowledge to realize strategic organizational value; it was also an early effort to articulate the aforementioned knowledge-based theory of the firm. Although this sampling strategy provides an incomplete representation of literature discussing KM, we reasoned that it would characterize those studies that have had the greatest impact on scholarly thought in MOS over this era—and is consistent with our goal of evaluating the approaches to knowledge management that have garnered the most scholarly attention. This choice allowed us to manage the challenge of reviewing a literature whose scope is well beyond what could be captured with any targeted effort.

We assembled our corpus using a Boolean search string in the Social Science Citation Index of Thompson Reuters' Web of Knowledge. For each year, we searched for articles whose title, abstract, or keywords contained any of the following key terms: "knowledge management," "knowledge sharing," "knowledge transfer," "knowledge differentiation," "knowledge integration," "knowledge process," "knowledge specialization," "knowledge flows," or "knowledge exchange." The outputs of these queries supported using 1996 as a start point as this year marked the outset of a nearly monotonic increasing trend in the amount of KM research published each year (see Online Appendix for article counts by year).[2] Next, we sorted each year's results by the number of times each article had been cited. We reviewed the top results for each year to ensure they expressed a primary interest in examining the processes by which knowledge flows within organizational contexts. We excluded articles that did not meet this criterion, replacing them with the next most cited article for that year. For example, we excluded multiple articles from the biological sciences that looked at knowledge processes as they relate to human evolution. We continued this method until we selected ten sources for each year. Our final corpus contained a stratified sample of 200 articles published over the past 20 years. The sample consisted primarily of empirical (126) and theoretical (65) articles with a smaller subset of reviews, simulations, and meta-analyses (9).

Next, we reviewed each of these articles to characterize how KM was represented therein; that is, how the research sought to discuss, represent, or analyze the integration or differentiation of knowledge. Each author read a subset of the total articles and engaged in a process of focused coding related to any mention of integration or differentiation of knowledge, then produced a short memo for each article reflecting the presence of knowledge integration or differentiation, and a longer memo summarizing the findings from the total group of articles. After an initial read of the literature, the authors met to discuss any emergent patterns or themes associated with views of KM represented in the articles. At this stage, we determined

[2] The Online Appendix is accessible at https://doi.org/10.5465/annals.2016.0041.

that characterizing studies through the binary of an integration or differentiation focus was too constraining and not reflective of how these knowledge processes were portrayed. Instead, KM integration and differentiation were presented as both (and simultaneously), outcomes and antecedents; they were also portrayed as both goals to be strived for and obstacles to be overcome.

To allow for a more nuanced characterization of the way knowledge processes were presented in the research, our next stage of analysis identified what we termed *knowledge trajectories*, which described the extent to which integration and differentiation were viewed as the starting or ending states for a specific knowledge management process. We chose the term "knowledge trajectory" to demonstrate an intended dynamic of change or stasis. Allowing for such temporality implies that organizational knowledge can be either maintained or transformed, such that organizational knowledge that is integrated at T_0 may either remain integrated or may become differentiated at T_1. We reviewed our initial coding of the articles and identified four knowledge trajectories present in the literature: integration to integration, differentiation to integration, integration to differentiation, and differentiation to differentiation. In settings where *integration to integration* is the focus, individuals, groups, or units start with common or shared knowledge and the goal is the continuation of this mutual knowledge over time. Studies addressing *differentiation to integration* discuss ways that organizations seek to bring together disparate knowledge such that it is then shared and accessible by others. Conversely, *integration to differentiation* notes ways organizations might start with shared knowledge and then individuals or subunits develop specializations or distinct forms of knowledge. Finally, *differentiation to differentiation* addresses efforts to sustain, maintain, or protect specialized knowledge such that it remains distinct and not common to everyone. It is important to note that these trajectories are not at all mutually exclusive; complex organizational systems are likely to include all four within and across their units (see Dooley & Van de Ven, 1999; Poole, 2014).

After identifying the four trajectories, we revisited our corpus and coded each article in terms of the knowledge trajectories discussed in the research. The Online appendix lists our sample of articles and final coding assignments. Table 1 below displays each trajectory in relation to the tension between integration and differentiation and labels them in terms of the distinct process of knowledge management supported by each trajectory: *Maintaining Common Ground*, *Producing Common Ground*, *Maintaining Specialization*, and *Producing Specialization*. We see this model as a valuable device for understanding the myriad processes examined by scholars interested in KM. In one sense, then, our model resembles Malone et al.'s (1999) efforts to create a "process handbook," a useful lexicon to grasp knowledge dynamics in organizing. Beyond its function as a vocabulary, however, it provides a meta-theoretical lens to capture and compare claims about trajectories in the substantial body of scholarship on knowledge and KM in MOS.

In the following sections, we outline the findings from research within each of the four quadrants of Table 1. We start by explaining the dominant scholarly focus on the differentiation to integration trajectory represented in the second quadrant of the table. We label this quadrant *Producing Common Ground* to demonstrate the goal of bringing together diverse knowledge into some shared form. We summarize research within this vein and then explain why this trajectory received disproportionate scholarly attention in our sample.

Next, we review research within each of the three other trajectories. As noted earlier, one benefit of focusing on knowledge trajectories is that it suggests that multiple KM processes exist not in fundamental opposition to one another, but rather as alternative

TABLE 1
Four Trajectories of Knowledge in Organizations and Their Prevalence in Our Sample (Count, % of Sample)

		T_1	
		Integrated Knowledge	**Differentiated Knowledge**
T_0	**Integrated Knowledge**	1. Maintaining common ground (16 papers, 8%) *Exemplar:* Internal boundary work (Aldrich, 1971)	4. Producing specialization (25 papers, 12.5%) *Exemplar:* Divisionalization (Galbraith, 1973)
	Differentiated Knowledge	2. Producing common ground (169 papers, 84.5%) *Exemplar:* Overcoming interpretive barriers (Dougherty, 1992)	3. Maintaining specialization (46 papers, 23%) *Exemplar:* Summarization & uncertainty absorption (March & Simon, 1958)

possibilities. Following Poole and Van de Ven's (1989) argument that acknowledging both sides of a theoretical paradox can be generative, we adopt the position that presumably oppositional forces, such as integration and differentiation, should be capable of coexisting dynamically with one another. Thus, it is important for us to maintain an awareness of how processes governing the three other trajectories play out in organizations. We demonstrate the value of this awareness by highlighting findings from research positioned in the remaining three quadrants of our model.

Differentiated to Integrated Knowledge: Producing Common Ground

Our analysis indicates that the most influential KM scholarship over the past two decades emphasized the study of integrational trajectories over others—with 84.5% of articles in the sample including this as a focus. The dominant focus on this trajectory seems to have emerged from the frequent characterization that the bringing together of disparate knowledge is a central mechanism by which organizations can realize strategic benefits. This emphasis can be traced to the earliest works in the sample. By way of illustration, Grant (1996a: 375) argued that "if the strategically most important resource of the firm is *knowledge*, and if knowledge resides in specialized form among individual organizational members, then the essence of organizational capability is the integration of individuals' specialized knowledge." From a resource management perspective, advantages were portrayed as coming not simply from the possession of unique knowledge, but from the ability to integrate available existing knowledge to achieve capabilities that no other firm possessed (Drucker, 1994; Nonaka, 1994). Topically, research in this quadrant examined interrelated phenomena at multiple levels of analysis.

Team-level studies have emphasized the interactional, structural, and relational mechanisms leading interdependent groups to effectively integrate individual-level knowledge into shared organizational knowledge (Bechky, 2003; Carlile, 2002; Gibson & Vermeulen, 2003; Levina & Vaast, 2005). Okhuysen and Eisenhardt (2002), for example, found that formal interventions could influence groups' capacities for sharing knowledge, showing that teams where members were prompted to interact more frequently experienced increased effectiveness at integrating their local knowledge in an analysis task. Levina and Vaast's (2005) study of boundary spanners at an insurance company and an internet consulting firm demonstrated the practices associated with brokering knowledge across intraorganizational boundaries, showing that only those individuals who actively negotiated relationships with diverse individuals realized the benefits associated with integrating diverse ways of knowing. At the heart of these studies, and others like them, is a recognition that teams that effectively integrate diverse forms of knowledge often realize measurable performance benefits (Carlile, 2004; Mesmer-Magnus & DeChurch, 2009).

Information technologies also played a central role in KM scholarship focusing on a differentiation to integration trajectory, with scholars quickly recognizing that such systems offer the potential to both centrally store codified knowledge and to connect individuals with experts with relevant tacit knowledge (Alavi & Leidner, 2001; Davenport, De Long, & Beers, 1998). Technology was promoted in KM scholarship as a way to provide broader access to common knowledge stores and ease relational development that would facilitate sharing of diverse knowledge. The belief that KM systems could offer beneficial integrative affordances remained despite evidence that early tools designed to become repositories of shared knowledge ultimately evolved into "information graveyards" with few contributors and fewer consumers (Bock, Zmud, Kim, & Lee, 2005; McDermott, 1999). Scholars argued that the integrative function of KM systems largely depends on motivating individuals to contribute to such systems (Chang & Chuang, 2011; Kankanhalli, Tan, & Wei, 2005; Lin, 2007). Wasko and Faraj (2005), for instance, showed that one of the major barriers to fostering integrative practices through KM systems arises because individuals tend to only contribute content when they perceive participation will provide them self-benefit. Without proper incentive, individuals tend to view participation in information systems as either wasteful or counterproductive (Vaast & Walsham, 2005). In recent years, studies of KM systems have shifted to examine the role that social media technologies may play in fostering knowledge integration in organizations (Leonardi, 2014; Majchrzak, Faraj, Kane, & Azad, 2013), noting that even though many of these technologies may fail to directly capture tacit knowledge, they can make knowledge of "who knows what" visible in ways that help individuals seek out others (Leonardi, 2007; Leonardi & Treem, 2012; Yates & Paquette, 2011).

Network studies have examined the structures associated with integrating local knowledge into organizational knowledge (Cummings, 2004; Fleming, Mingo, & Chen, 2007; Hansen, Mors, & Løvås, 2005; Inkpen & Tsang, 2005; Singh, 2005). A recurrent finding in this work is that actors who are positioned in structures offering access to diverse knowledge tend to exhibit comparatively higher performance with respect to idea generation (Burt, 2004), the novelty of their inventions (Fleming et al., 2007), or the impact of scholarly outputs (Uzzi, Mukherjee, Stringer, & Jones, 2013). By connecting multiple bodies (and practices) of knowledge, such individuals seem capable of integrating disparate knowledge.

At a macro level, studies have examined the factors leading to knowledge transfer in alliances and joint ventures (Ireland, Hitt, & Vaidyanath, 2002; Oxley & Sampson, 2004; Sampson, 2007). We see an integrational motivation here again, with such ventures being characterized as opportunities for an organization to access and use its collaborators' knowledge to its own benefits. This line of scholarship recognizes that firms and groups differ in their abilities to capture, integrate, and apply knowledge from diverse sources (Cohen & Levinthal, 1990; Volberda, Foss, & Lyles, 2010).

Explaining the integration bias. Our analysis uncovered several explanations for why this integrational emphasis materialized. Specifically, we argue that conceptualizations of tacit knowledge as a barrier to performance, a perspective viewing value as produced in the (re)combination of previously distributed resources and an emphasis on meeting organizations' perceived operational needs, have all driven this trend.

Viewing tacit knowledge as a barrier to KM. The existence of tacit knowledge has emerged as one of the key impediments to realizing a central tenet of effective KM in organizations: "knowing what we know." Polanyi (1961, 1962, 1966) introduced the term tacit knowledge to recognize that not all knowledge is created equal, nor is it manifest in similar ways. Whereas some forms of knowledge are easily codified and transferred, others are embedded within experience and practice. This distinction reveals a fundamental tension for organizations hoping to derive value from perceived knowledge resources. If knowledge is always tied to practice, then it is difficult for that knowledge to exist outside of a finite context or the experience of individuals immersed in practice. As such, research often characterizes tacit knowledge as the "problem" requiring solution (Cramton, 2001; Sole & Edmondson, 2002; Zack, 1999). KM scholars recognized that tacit knowledge was among the chief factors leading to knowledge being "sticky" inside organizations (Brown & Duguid, 2001; Dougherty, 1992; Osterloh & Frey, 2000; Szulanski, 1996, 2000) and, therefore, difficult to move or extract.

The challenge of overcoming the stickiness of tacit knowledge is routinely invoked as a rationale for explicitly focusing on integrational knowledge trajectories. KM research has frequently addressed mechanisms by which tacit knowledge may be transferred to others or converted into codified forms. For instance, collaboration and knowledge sharing research has examined how differing communities can share knowledge across boundaries of practice (Carlile, 2002; Wenger & Snyder, 2000), largely revealing that sharing tacit knowledge requires active engagement and cross-boundary dialogue (Amin & Roberts, 2008; Carlile, 2004; Levina & Vaast, 2005; Wang & Noe, 2010). This engagement often takes place in communities of practice (Brown & Duguid, 1991; Iverson & McPhee, 2002) where individuals engage in similar tasks and roles, and can share knowledge, observe each other, and learn the tacit knowledge needed to operate as a competent member of an organization. Others have considered ways that individual tacit knowledge can be transformed or converted into a more explicit form and made available to others in an organization (Nonaka, 1994). This goal of externalizing tacit knowledge is a key driver of the implementation of KM technologies intended to facilitate broader availability of individual-level knowledge as a resource available across the firm. However, research on KM systems has shown that traditional strategies for capturing and cataloging knowledge are often ineffective for sharing tacit knowledge because it is difficult for individuals to externalize this knowledge in a way that it would be contextually useful (Ackerman, Dachtera, Pipek, & Wulf, 2013; Huysman & de Wit, 2004; McDermott, 1999).

Viewing the (re)combination of knowledge as a source of innovation. Another theme that emerged was a prevailing view of the (re)combination of preexisting knowledge as a fundamental source of organizational benefit (Gertler, 2003). Specifically, integrating diverse knowledge resources is often seen as a significant predictor of the likelihood of innovative outcomes. This viewpoint descends largely from Schumpeter's (1934) recognition that competitive advantages are often gained through the application of previously existing capabilities in new contexts. This perspective toward knowledge and innovation is readily visible in definitions of knowledge creation throughout the literature such as the following from Inkpen and Dinur:

> Clearly, organizations are repositories of knowledge. The important question is how individual and group interactions contribute to organizational knowledge creation. Organizations cannot create knowledge without individuals, but unless individual knowledge is shared with other individuals and groups, the knowledge will have a limited impact on organizational effectiveness. Hence, organizational knowledge creation should be viewed as a process whereby the knowledge held by individuals is amplified and internalized as part of the organization's knowledge base. (1998: 456)

This definition implies that the key problem organizations face is not one of creating *new* knowledge (in the sense of creating something that *never* existed previously), nor is the problem one of maintaining knowledge diversity across a workforce. Rather, the issue to be managed is one of integrating preexisting individual-level knowledge to realize unrecognized organizational benefits. Research in our sample emphasized the processes by which organizations connect and foster knowledge exchange from diverse sources. Practice-oriented research showed that teams taking explicit efforts to connect with others produce more effective innovations (Hargadon, 2003; Hargadon & Sutton, 1997), and network-oriented research demonstrated that bridging multiple information sources increases the propensity for individuals or organizations to produce impactful outputs (Breschi & Lissoni, 2009; Burt, 2004; Fleming & Singh, 2010; Phelps, 2010)

Diverse knowledge is often treated as the initial condition upon which studies of knowledge in organizations begin. Relatedly, knowledge asymmetry is characterized as a consequence of the increasing prevalence of white collar work that favors specialized or esoteric work (Drucker, 1994; Sharma, 1997), and theories of group interaction assume that individuals will divide responsibilities for knowledge domains and related tasks over time (Brandon & Hollingshead, 2004; Wegner, 1987). These perspectives assume a natural entropy toward the specialization and separation of knowledge among organizational members, and implicitly or explicitly position this as a challenge to be addressed (Argote & Ingram, 2000).

Measurable operational benefits from integration. Of the four trajectories, integrational processes are the most clearly linked to measurable performance benefits, especially at short temporal spans. These measurable benefits are likely one of the strongest reasons this trajectory has emerged as the dominant object of inquiry for KM, particularly for scholars in strategic management who seek to configure systems for maximum efficiency and effectiveness (Gold, Malhotra, & Segars, 2001; Zheng, Yang, & McLean, 2010) Alhough integrated knowledge is presented as promising operational benefits, many firms consider integration to be the KM trajectory that provides the most difficulties. This was particularly true during the era when the KM field was first developing. Firms in the 1990s were rooted in the shift to a postindustrial economy (Drucker, 1994), globalization was a growing phenomenon (Friedman, 1990), and information technology was beginning to provide organizations with access to sources of knowledge beyond previous comprehension. The issue at hand was not one of *creating* something new, but of taking advantage of the vast resources sitting there for the taking. Titles of a number of early KM articles such as "If only we knew what we know" (O'Dell & Grayson, 1998) and "Embedded firms, embedded knowledge" (Lam, 1997) belie this viewpoint. Ruggles (1998) surveyed 431 CEOs and showed that the leadership of most organizations viewed integration as their primary problem. One striking finding from Ruggles' analysis was that the least prominent issue reported by respondents was *creating new knowledge.* Rather, CEOs saw the problems of managing knowledge as primarily including issues of transferring, capturing, and embedding preexisting knowledge documentation. As Alavi and Leidner (2001: 108) argued, "It is less the knowledge existing at any given time per se than the firm's ability to effectively apply the existing knowledge to create new knowledge and to take action that forms the basis for achieving competitive advantage from knowledge-based assets."

Another influence producing the focus on integration is the tendency for KM scholarship to emphasize knowledge outputs, such as the production, addition, or combination of knowledge that are accessible to multiple organizational actors. In many respects, differentiation is characterized as the *absence* of shared knowledge outputs. Adopting an integrational focus allows scholars to analyze how strategic KM initiatives alter performance in organizational contexts (Dyer & Nobeoka, 2000). This is done both within organizations related to how teams and groups integrate knowledge, as well as at the organizational level related to the formation of organizational alliances (Lam, 1997; Lane, Salk, & Lyles, 2001; Zucker, Darby, & Armstrong, 1998). A review of KM literature in the field of information systems, for instance, found that a clear majority of studies had adopted a normative focus on the ways the presence and use of KM technologies influenced the performance of organizations, individuals, or units (Schultze & Leidner, 2002). In sum, because the use of KM in organizational systems has been framed as a means to enhance organizational performance by capitalizing on already-existing knowledge, a concern for how to foster knowledge integration (and a concomitant inattention to knowledge differentiation and the creation of new knowledge) was the understandable scholarly trajectory.

Calls to consider alternative trajectories in KM research. Although integrational research dominated our sample, a parallel set of theorists have argued for more dynamic perspectives on KM (Venkitachalam & Willmott, 2017). Early review articles emphasized that integration was only one component of a complete KM strategy, recognizing that knowledge creation and knowledge protection also played important roles in the dynamics of knowledge within and between organizations (Demarest, 1997; Wiig, 1997). Bontis (1999: 436) argued that the concept of knowledge creation, which he conceptualized as the discovery and production of differentiated knowledge, had been "virtually neglected even though Nonaka and Takeuchi (1995) are convinced that this process has been the most important source of international competitiveness for some time." De Long and Fahey (2000) argued that one of the reasons that many KM efforts were met with failure was due to an overemphasis on processes of codification and transfer that lacked recognition for the dynamic nature of knowledge in organizations. Tsoukas, too, has long criticized KM research for failing to account for the importance of locally differentiated and performed knowledge in day-to-day work practices (Tsoukas, 2009; Tsoukas & Vladimirou, 2001). Others have argued that KM scholarship has paid little attention to mechanisms by which organizations create *new* knowledge across and within different organizational units (Argote & Miron-Spektor, 2011) and claimed that terms like "knowledge creation" are often treated as an outcome of integrating previously isolated bodies of knowledge as opposed to the generation of something novel (Phelps, 2010).

Our four-trajectory framework offers a conceptual lens with which to characterize these critiques. Although operational benefits may readily emerge from developing KM processes emphasizing the production of common ground, focusing solely on those processes overlooks the important role that the other three trajectories play within organizational systems. Our review suggests that although the other three trajectories are less frequently represented in the literature, they nonetheless encompass organizational processes that are vital to sustaining ongoing organizational performance. We now turn to research in the remaining quadrants of our model to illuminate the value of these alternate processes.

Integrated to Integrated Knowledge: Maintaining Common Ground

The trajectory captured in the upper-left quadrant of Table 1 portrays KM as the practice of sustaining integrated knowledge over time. We labeled this quadrant *Maintaining Common Ground* because it describes processes in which organizational units begin with a body of shared knowledge and enact problem-focused action to preserve that communal knowledge. This trajectory was least represented in our corpus, appearing in just 8% of our sample, and is largely present in studies that discuss the maintenance or perseverance of partnerships (Beckman, Haunschild, & Phillips, 2004; Malhotra, Gosain, & El Sawy, 2005), relationships (Im & Rai, 2008; McFadyen & Cannella, 2004), or collaborations (Fleming, Mingo, & Chen, 2007). Maintaining common ground is viewed as a means of retaining or protecting existing knowledge resources available among multiple actors.

There are several potential organizational benefits associated with maintaining common ground. Looking beyond the most influential articles, we can see how this knowledge trajectory can play important roles in fostering coordination across multiple subunits or domains (Okhuysen & Bechky, 2009), facilitating effective decision making (Cohen, March, & Olsen, 1976; Padgett, 1980b), and managing organizational identities and boundaries (Aldrich, 1971). A few illustrations will help demonstrate the value associated with this type of process. First, Bruns's (2013) ethnographic account of systems biologists engaged in cancer research located a variety of practices enabling team-level coordination and collaboration. Although these scientists hailed from multiple disciplines, they took explicit efforts to maintain a common interpretation about their tasks. Maintaining this interpretation allowed the scientists to see their efforts as aligning with an overarching goal, even when working apart from one another. Actively managing this shared knowledge facilitated the complex coordination necessary to facilitate an effective collaborative relationship over time. Second, Bechky's (2006) study of role-based coordination in film projects showed how a generalized role structure traveled across projects, producing the coordination and continuity that enabled each ad hoc system to materialize. Role structures are not, however, static and preexisting objects existing apart from practice; instead, they are interactively (re)constituted within and across projects through distinct practices such as interpersonal thanking, admonishing, and joking. Both studies demonstrate how maintaining a relatively small set of common epistemic objects across subunit boundaries (Rheinberger, 1997) can facilitate commensurate action in contexts that would otherwise break down into conflict or chaos. The common ground maintained in these studies was not a shared set of information or beliefs about some object, but an understanding of roles in a system comprising heterogeneous roles.

A similar claim is advanced in Huber and Lewis's (2010) theoretical discussion of "cross-understanding": group members' grasp of other members' mental models. This concept is associated not with a division of cognitive labor, but with collaboration, the surfacing of task-relevant information, and member learning—all of which, in turn, shape group performance. Groups can maintain common ground through shared meta-knowledge regarding who knows what. In this situation, which is represented in transactive memory systems, the organization or group strives to construct a shared directory of the knowledge of organizational members and seeks to update and maintain that directory as new knowledge enters the system or the group demands change (Lewis, Lange, & Gillis, 2005; Wegner, 1987). Importantly, maintaining this common meta-knowledge requires ongoing efforts in the form of group interaction and communication (Hollingshead, Brandon, Yoon, & Gupta, 2010). Cross-understanding, such as the empirical cases presented in the preceding paragraph, suggest that *Maintaining Common Ground* is not about replicating knowledge, as would be the case if integration meant that each individual's cognitive content becomes isomorphic by T_1. Instead, when the unit of analysis is a collective—a team, group, project, or organization—this trajectory highlights how that collective's task performance *reinforces* its approach to generating harmonized action. Knowledge of the task, of the role structure, of others' knowledge, and of how to generate coordination all remain integrated as a result of conjoint activity.

Differentiated to Differentiated Knowledge: Maintaining Specialization

The third quadrant contains a trajectory we call *Maintaining Specialization*, which was present in nearly one-fourth (23%) of all articles and was the second most prevalent trajectory after *Producing Common Ground*. This movement starts with a condition of differentiated knowledge but, instead of creating integrated knowledge or common ground, upholds the divergent knowledge characterizing the organizational unit. Organizational theory has long argued that such processes are logically necessary: If organizations require specialization to interact with, and adapt to, complex environments, KM processes should not be expected (or intended) to result in integrated knowledge in every case (March & Olsen, 1976; Padgett, 1980a, 1980b). *Maintaining Specialization* speaks to the desire to retain differentiation across tasks and time. Differentiation allows organizations to retain a wider array of knowledge (Bock, Zmud, Kim, & Lee, 2005; Lavie, Steiner, & Tushman, 2010) and to protect distinct forms of knowledge providing them competitive advantage (Oxley & Sampson, 2004). Alternately, maintaining specialization may be advantageous when the required work to maintain common ground among groups is too costly (Carlile, 2002).

A body of work that demonstrates this process is found in the motivated information sharing literature. This work acknowledges that the tasks facing groups frequently elicit mixed (and even conflicting) motivations among group members, leading them to pursue varied approaches to search for, and process, information. Group discussions, particularly when working on a hidden profile task (where some information is shared and some is unshared by group members, and no member can determine the best group decision before the interactive pooling of information), tend to elicit shared information to the detriment of the unique information possessed by individuals (Stasser & Titus, 1985). Studies following Stasser and Titus often depict motivated information sharing as a problem to be overcome; when the aim is to improve decision-making in this specific type of group task or to achieve the ends proffered by knowledge integration, a lack of member sharing may indeed be a concern. Some research, however, acknowledges that maintaining differentiation in group members' information can be functional for other (non-hidden profile) tasks and may serve group members' varying motivations (Brodbeck, Kerschreiter, Mojzisch, Frey, & Schulz-Hardt, 2002; Steinel, Utz, & Koning, 2010; Wittenbaum, Hollingshead, & Botero, 2004). Moreover, under certain conditions, groups that maintain specialization with respect to information may experience *increased* ideation and creativity (Bechtoldt, De Dreu, Nijstad, & Choi, 2010). Motivated information sharing research thus recognizes that maintaining specialized knowledge through the withholding of information is an active process that has consequences for group performance. These processes begin with, and maintain, divergent knowledge; the point is not that group processes mold this into shared knowledge, but that diversity, managed well, can often produce superior decision-making processes.

A second illustration of *Maintaining Specialization* comes from a rather different line of work. Bruni, Gherardi, and Parolin's (2007) explicitly practice-based perspective (see also Brown & Duguid, 1991; Cook & Brown, 1999) positions their view against conventional visions of knowledge as an item to be transferred:

> When we conceive knowledge as a substance, we see it materialized in objects; when we conceive it as a property, we see it as owned by individuals. When we look at knowing-in-practice, we define it as the mobilization of the knowledge embedded in humans and nonhumans performing workplace practices. (p. 85)

From this stance, Bruni et al. studied remote health-care consultations, finding a system of *fragmented* knowledge. Knowledge, in this setting, was embedded in patients, the medical community, organizational rules and habits, artifacts, a technological infrastructure, and an organization; no single site eclipsed the others. The authors located practices that sometimes created a temporary alignment of actors in the conduct of remote consultations, which also could lead to the disruption and disturbance of practices, roles, and knowledge. Their point is that the *multiplicity* of heterogeneous elements must be mobilized and transformed in situated practice; the elements are related to one another, but the fragmented system never experiences complete integration (see also Gherardi & Nicolini, 2000; Hopwood, 2014). Research operating from this perspective must remain sensitive to the contingencies and practices through which temporary alignments—but not integrated knowledge—occur. This claim about the sensitivity to the contours of situations can also be seen in Faraj and Xiao's (2006) study of a medical trauma center, where teams employed different forms of coordination, based on expertise or dialogue, to address exigencies of the urgent situations. Across these two lines of thought on *Maintaining Specialization*, then, are claims that processes that retain or re-create differentiated knowledge can be functional for organizational practice.

Integrated to Differentiated Knowledge: Producing Specialization

Our final quadrant encapsulates a trajectory that is represented relatively rarely (in 12.5% of the sample) in the KM literature: *Producing Specialization*. Here, units begin with shared knowledge but, through particular processes, transform that integration into a modicum of differentiation. Although the production of specialization may at first glance seem to be counter to the common goals of KM, some research suggests that this type of process plays an important role in knowledge systems over time. Empirically, this is represented in work that shows the value of firms developing specialized knowledge that allow them to act more efficiently in a variety of domains (Brusoni, Prencipe, & Pavitt, 2001; Hansen, 2002). Knowledge may be available in a shared or communal form and then be applied or developed by individuals or organizational units in novel ways to pursue specialized tasks (Majchrzak, et al., 2013; Tsoukas, 2009).

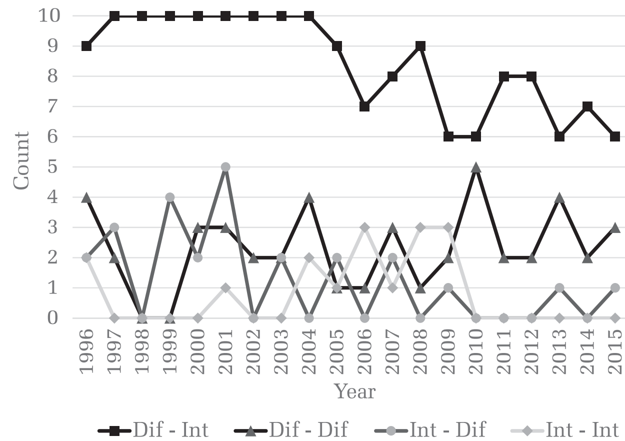
One perspective on this process draws on differences between individuals and the systems in which they operate, portraying the increased specialization and division of labor across individuals as creating greater productive capacity for groups (Austin, 2003) and firms (Kogut & Zander, 1996). As individuals or units differentiate, the span of the environment that the organizational system, writ large, covers increases. In some work, the production of specialization is tied to the creation of new knowledge, as in DeLong and Fahey's (2000) conceptual argument about the importance of cultures that encourage debate and internal interrogation of existing assumptions, moves that can disrupt previously shared understandings but can, in time, create novel courses of action. At the level of interorganizational networks, the value of creating knowledge through specialization can be seen in Brusoni, Prencipe, and Pavitt's (2001) study of managers and designers in the aircraft engine control field. Noting that a division of labor and a division of knowledge refer to separate concepts, these authors suggested that, within networks of loosely coupled organizations, there are benefits to encouraging firms to develop more knowledge than that which is minimally necessary for their contribution to the product; this allows those firms to play a coordination role in the network.

An alternative perspective on *Producing Specialization* is evident in research that foregrounds the process of forgetting as a component of organizational learning. De Holan and Phillips (2004a, 2004b) explicate this view, arguing against the MOS literature's almost singular focus on the accumulation of new knowledge in defining organizational learning. They note that organizational memories can be hindrances, rather than aids, to organizational performance (Anand, Manz, & Glick, 1998; Crossan, Lane, & White, 1999) and that advantages based on memories are likely to be more precarious than those based on more stable resources. The process of forgetting may be involuntary, a result of human limitations or a systemic inability to codify knowledge (Argote, 2012; Nonaka, 1994). Yet it is also often intentional, involving the construction of barriers that prevent past information from being included in ongoing routines, or it can consist of actively deleting or destroying elements of the present to clean the slate for the future (Bowker, 1997). Forgetting can, consequently, be functional in that it can lead to the creation of new routines and new forms of specialization that aid in ongoing action.

Summary

Our review of influential KM research over the past 20 years shows a disproportionate focus on a knowledge trajectory of differentiation to integration, yet the minority of literature recognizing other knowledge trajectories note a variety of potential benefits these divergent efforts support and the ways multiple processes of knowledge are key to successful KM. Figure 1 illustrates the number of articles that include each trajectory as an object of their discussion for each year in our sample. Our coding scheme allowed for articles to represent multiple processes; however, most articles (118) exclusively studied integrational trajectories without discussing other mechanisms. This pattern was emphasized in empirical articles (with 85/126 explicitly on *Producing Common Ground*) and deemphasized in theoretical work (31/65). The downward slope in the integration line over the past decade potentially indicates an ongoing shift in scholarly thought toward a more representative mix in objects of analysis.

FIGURE 1
Count of articles representing the four trajectories by year



It is important to recognize that our conclusions regarding the integration bias are bounded by the scope of our review. Our goal was not to assess the breadth of KM scholarship across all outlets and disciplines; instead, we sought to gauge what has garnered to most attention within MOS. This alongside our sampling method mean that we can only conclude the emphasis on integration has been present within the most highly cited articles on KM in MOS. A vast array of studies beyond this sample have examined KM and many of these have emphasized trajectories beyond the production of common ground. In what follows, we build on such studies, looking beyond our corpus to lay out what we see as three fruitful directions forward for KM scholarship.

TRANSCENDING THE INTEGRATION BIAS: STRATEGIC DIRECTIONS FOR KM RESEARCH

Given the presence of the integration bias in studies of leading KM scholarship, our task now turns to considering how MOS scholarship might build on the four trajectories to produce novel insights into organizations and organizing processes. Recognizing that the field's emphasis on integration has emerged from both a strategic and a practical motivation, any argument for the value of a more dynamic viewpoint on KM must preserve the operational goals that have driven MOS scholars' interest in knowledge for the past two decades. It would not be enough for us to simply claim that a dynamic approach to knowing in organizations offers a more ecologically valid characterization of how firms operate. Rather, we need an agenda of research that will demonstrate *how* considering knowledge differentiation in organizations can offer advantages over continuing to emphasize integration over other dynamics.

In this final section, we outline three different directions for ongoing research that we see as avenues for simultaneously broadening and reconciling the range of KM scholarship. Each approach explores how taking a dynamic perspective on knowledge and knowing can not only engage fruitfully with the complexity of knowledge, but also offer new strategic directions for KM research. Moreover, each approach argues for the benefit of incorporating multiple trajectories as analysts' objects of analysis. First, we argue for a perspective considering how organizations can be mindful of *which* knowledge they share by considering the contexts where refraining from information sharing might offer organizations benefit. Next, we build an argument for how taking a situated and dynamic view of knowing can be applied to an organization's benefit, and how doing so can offer pragmatic and action-oriented guidance for a variety of organizational types and contexts. Finally, we argue that viewing KM as processes of continuous organizational change can aid researchers in maintaining a focus on opposing forces and multiple trajectories of knowledge over time. Table 2 summarizes the research questions that become centralized by each of these approaches.

TABLE 2
Three Strategic Directions for KM Research

Approach	Research Direction	Related Research Questions
Acknowledging the value of uncommon ground	Evaluating which knowledge to share	How do experts determine what *not* to share in a particular opportunity for engagement? How can organizations support, reify, and codify these processes? Under what conditions is the development of interactional expertise operationally efficient?
Investigating practices of differentiated knowing	Uncovering "invisible" work in KM	How does the invisible interactive work conducted by human and nonhuman participants develop and promulgate assessments (i.e., metrics and indicators) of knowledge and knowledge work? How, through those interactions, do particular interests become inscribed into those assessments? Are particular forms of practice associated with an emphasis on either common ground or specialization in KM?
	Heterogeneity in knowledge and knowing	How does knowledge/knowing heterogeneity produce practices that reframe and/or challenge existing KM initiatives? How do efforts to engage heterogeneity foster or inhibit the innovative management of integration–differentiation tensions?
Viewing KM as a process of organizational change	Temporality in KM as organizational change	How do actors' interpretations and uses of KM tools shift over time? How do changes in interpretations and uses of KM tools influence processes of knowledge integration or differentiation?
	Knowledge networks	Do differential types of actors (human and nonhuman) or relations influence processes of knowledge integration and differentiation in different ways? How does knowledge integration or differentiation at one level of an organizational network influence KM processes at another level of the network? How does the articulation of an organizational knowledge network, and associated meta-knowledge, provided by KM efforts influence processes of knowledge integration or differentiation?

Acknowledging the Value of Uncommon Ground in (Not) Sharing Knowledge

Our first avenue forward begins from the premise that social processes that facilitate differentiation (the trajectories we called *Producing Specialization* and *Maintaining Specialization*) can play an important role in enabling the beneficial impacts of integrating diverse knowledge. Acknowledging differentiation is not to disparage the value of integration, but is a recognition that knowledge systems governed solely by integrational dynamics will quickly find themselves disadvantaged. Specifically, we argue that recognizing the important role of *uncommon ground* in KM requires us to ask less about how organizational units integrate knowledge and more about *which* knowledge *should* be integrated across organizational units. Such an analytic move may be conceptualized as building research spanning the left and right halves of Table 1.

The Carnegie School offers a point of departure for appreciating the reciprocal roles of differentiation and integration in KM systems. In this body of theory, organizational structures and practices function to facilitate decision-making activities in a constantly shifting knowledge environment whose scale is beyond rational conception (Cyert & March, 1992; Gavetti, Levinthal, & Ocasio, 2007; March & Olsen, 1976; March & Simon, 1958; Simon, 1997). At the same time, each organizational unit is characterized by bounded rationality, which means each is limited in the scope to which it may attend at any moment in time. Given these constraints, one reason for the firm's existence is its capacity to enable attention to a wider swath of the environment than possible by any single individual (or unit). By dividing into specialized subunits, firms spread their knowledge and create the potential for complex, efficient, and coordinated action. For the Carnegie School, differentiation and integration play different but equally important roles in enabling organizational action: differentiation facilitates exploratory processes which develop and expand the bounds of an organization's knowledge (March, 1991; March & Simon, 1958), and integration enables an organization to benefit from the knowledge it already possesses (Burt, 2004; Hayek, 1945; Ocasio, 1997; Padgett, 1980a, 1980b).

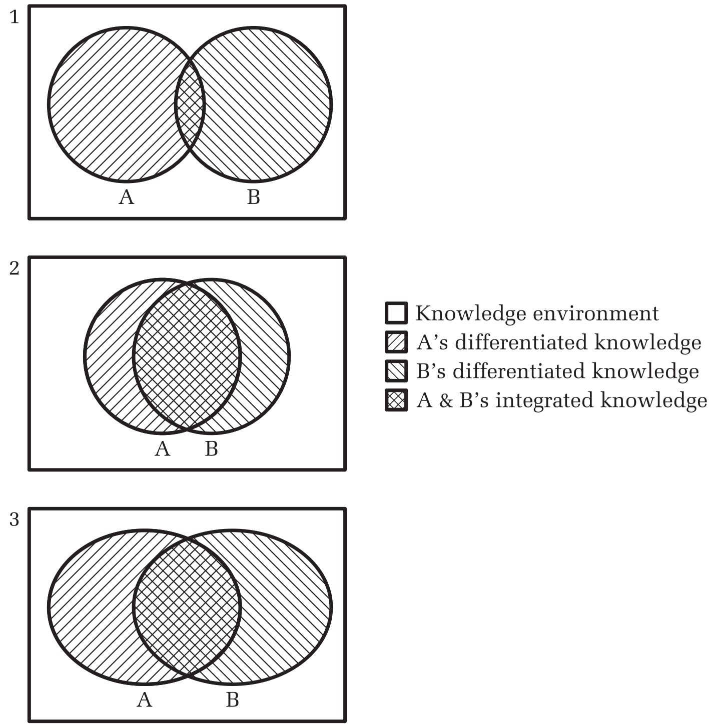
In a series of simulation models, March (1991) demonstrated how a disproportionate emphasis on either trajectory (in his terminology, exploitation or exploration) would result in the organization sinking into a local minimum in terms of performance. March's claim is echoed by Nonaka (1994), who held that effective KM involves managing the delicate balance between redundant knowledge and the requisite variety of subunits' knowledge necessary to foster generative interactions. Grant (1996b) too recognized that common ground played an important role in fostering communication across boundaries, but argued that "the key to efficiency is to achieve affective integration while minimizing knowledge transfer through cross-learning by organizational members" (p. 114). In short, then, effective KM requires organizations to enact coordination mechanisms that manage this tension and preserve the trajectories of both differentiation *and* integration.

Figure 2 illustrates this tension by showing how, when two organizational units interact, efforts favoring the development of integrated knowledge can produce the unintended consequence of minimizing the breadth of knowledge captured within the organizational system. Although the first two panels demonstrate how each unit's differentiated knowledge shrinks as overlap is produced, these panels also assume each unit maintains awareness of a consistent amount of knowledge in its environment over time.

The third panel, by contrast, visualizes how organizational units might *simultaneously* cultivate common ground and specialization Managing the tension between these impulses is likely to require a good deal of additional effort (and resources) because each subunit must be aware of a larger portion of the knowledge environment (hence the expansion from circles to ovals in the Figure). In other words, avoiding the sacrifice illustrated in panel 2 (ignoring the knowledge environment in the interest of achieving greater overlap with the other unit) requires the ongoing enactment of differentiation *in addition to* integrational processes. Managing the tension requires work; any effective KM strategy must acknowledge that labor and support it.

What, then, are the processes that foster the organizational management of the integration– differentiation tension? Although rarely using the notion of KM, research in MOS has addressed this question. For example, Hutchins (1995), in an ethnographic study of navigation teams on aircraft carriers, showed how artifacts and routines can allow individuals to appropriate external knowledge without having to develop situated understandings in a new domain. A technician may be trained to effectively adopt an astrolabe without developing deep familiarity with the underlying mathematical and astronomical knowledge that went into building such a tool. By taking advantage of the

FIGURE 2
The Tension between Differentiated and Integrated Knowledge.



Note: Panel 1 illustrates two organizational units with highly differentiated knowledge. Note that the total area of the environment they cover is quite large. Panel 2 illustrates two units with a large amount of integrated knowledge at the cost of conceptual breadth. Put formally: As $A \cap B$ increases, $A \cup B$ decreases. Panel 3 illustrates a contingency where the two units increase integration while preserving differentiated knowledge. Note how doing so requires each subunit to maintain awareness of a larger area of its knowledge environment.

knowledge embedded within the tool, the individual is free to devote his or her attention to other efforts involved in navigating the ship. In another example, Majchrzak, More, and Faraj (2011) explored the practices by which three cross-functional teams coordinated knowledge—and simultaneously avoided deep understandings of each other's area of specialty. Each team sought to develop just enough common ground to facilitate action, but did not engage in the deep dialogue that much of the literature (Bechky, 2003; Carlile, 2004; Kellogg, Orlikowski, & Yates, 2006) has emphasized as important to developing collaborations. In fact, avoiding deep dialogue helped teams side-step potential conflicts that would have emerged had they sought to achieve common knowledge. Furthermore, Bechky's (2006) study of role-based coordination in film production teams demonstrates a similar mechanism by which knowledge can be coordinated without actively engaging in integration-oriented dialogue. She found that culturally engrained understandings of the *roles* that occupational groups play in the film production process allowed for rapid assembly and execution. The three techniques—embedding knowledge in artifacts and routines, avoiding deep dialogue and operating on broadly accepted conceptions of roles—demonstrate that managing the integration–differentiation tension is a complex accomplishment, one that often relies on shallow (rather than deep) knowledge of both the task and of others' roles. At the same time, however, these studies demonstrate that maintaining differentiation is *not* a passive process: Routines need to be established and learned, groups need to develop norms of trust that allow for minimal sharing, and occupations need to develop the role-based knowledge necessary to afford this type of engagement. If KM scholarship were to pursue a deeper understanding of this type of balance, we see at least two areas worthy of increased scholarly attention: producing a deeper understanding of the skills involved in determining which knowledge to share and understanding the potential unintended consequences associated with integrating knowledge.

An important direction for KM scholarship in grappling with the tension between integration and differentiation is to examine how organizational units evaluate which knowledge *needs* to be shared in a context. If there is value to being judicious about which knowledge is worth sharing, then *evaluation* will be central to realizing organizational benefits associated with integration and differentiation. Along these lines, sociologists of expertise, such as Harry Collins (Collins, 2013; Collins & Evans, 2002, 2007; Collins, Evans, & Gorman, 2007), have written about the growing importance of what they call *interactional expertise* in contemporary organizational contexts. Interactional expertise is a mastery of the language of a particular domain absent the capacity to contribute to that domain's practice; it plays an important role in facilitating the coordination and translation of knowledge across boundaries (Collins & Evans, 2007).

Individuals with interactional expertise in another individual's or group's domain of practice have a greater capacity to translate coded activities across existing boundaries, aid in the resolution of conflicts, and facilitate actors' senses of mutual accountability to the practice. Importantly, because this expertise is based around discussion of practice, and not the material and embodied abilities defining practice, it may be easier and less costly to acquire, and accessible to a wider group of individuals (Collins et al., 2007). The recent attention to the importance of in-house *organizational anthropologists* (or ethnographers) resonates with this move (Tian & Walle, 2009). Although some organizational anthropologists are charged with understanding customer beliefs and behaviors, the role is more frequently about understanding organizational practice and generating insights for use in strategic decision-making (Jordan, 2010); some even argue that strategic managers should inhabit this role to develop a sensitivity to the cultures and practices characterizing their organizations (Ruben, DeNisi, & Gigliotti, 2017). Provided proper authority, organizational anthropologists can also foment dissent and interrogation of assumptions alongside efforts to create shared knowledge, based on the idea that it is only from *within* practice that judgments about whether (and how) to emphasize either integration or differentiation can be fruitful.

When individuals—whether or not they are organizational anthropologists—possess interactional expertise, they are capable of framing their communication with cross-boundary peers in a manner that fosters exchange without producing overlap. More importantly, interactional expertise can play an important role in allowing individuals to gauge precisely how much knowledge needs to be shared in a specific moment (Barley, 2015; Nonaka & Toyama, 2007; Shotter & Tsoukas, 2014; Treem & Barley, 2016). In MOS research, concepts such as cross-understanding (Huber & Lewis, 2010), directory development (Brandon & Hollingshead, 2004; Yuan, Fulk, Monge, & Contractor, 2010), and meta-expertise (Faraj & Sproull, 2000) describe similar dynamics, and point to the increasing importance that this type of understanding of an organization's subunits plays in affording the judicious balance between integration and differentiation.

Practically understanding these tensions in action will require a shift in the types of research questions that KM scholars ask as they approach contexts of organizing. Rather than seeking to understand the mechanisms by which organizational units share their knowledge, investigators might instead ask *How do interactional experts determine what not to share in a particular opportunity for engagement? And, moving beyond individuals' capacities, how can organizations support, reify, and codify these processes?* However, in keeping with our framing of KM as a complex and precarious accomplishment, it is important to acknowledge interactional expertise as a radically contextual and relational construct, which implies that developing it at one cross-unit boundary does not necessarily imply an ability to interact judiciously across all others (Collins & Evans, 2007). Given this limited transferability, it will also be important to gain insight into *the contexts under which the marginal returns on performance associated with developing interactional expertise are greater than the costs associated with developing it.*

Investigating Practices of Differentiated Knowing

Our analysis of the KM literature revealed not simply a bias toward integration, but also an assumption about the character of that which is being managed. The standard view in the KM literature understands knowledge to be "an identifiable entity, a commodity stored (i.e., located) *in* brains or bodies" (Kuhn, 2014: 482. emphasis in original). Such a vision of knowledge, argue Suddaby and Greenwood (2001), is by no means natural; instead, it is the result of the deployment of particular assumptions and sustained intellectual effort. This standard view, which Suddaby and Greenwood call *commodification*, occurs when knowledge "is abstracted from context and reduced to a transparent and generic format that can be more easily leveraged" (p. 934). As discussed in our section exploring explanations for the integration bias, when the principal aim of KM is to produce either common ground or specialization (as in Table 1) in the service of (a) accumulating, extracting, transferring, sharing, and protecting knowledge; (b) embedding knowledge in routines; or (c) capitalizing on knowledge via exploitation, then conceiving of it as a commodity, an *object*, appears logical.

This epistemological grounding framing knowledge as an object, however, has served as a point of contention since the advent of KM as a scholarly concern. Critics have drawn on a long line of social theory to denounce this cognitivism inherent in commodification at both the individual and collective level. Scholars have argued that the knowledge we see as possessed by an actor is usually less stable, transferrable, and replicable than assumed (Winter & Szulanski, 2001). Detailed understandings of the *doing* of work expose how knowledge is not simply a detached input into, or inhabitant of, organizational routines; it is an outcome of struggles over meaning and histories of activity. Knowledge should, therefore, be understood not as an object or a commodity, but as an always-precarious product shaped by the languages in use—one that must be continually re-accomplished in situated activity (Antonacopoulou, 2008; Brown & Duguid, 2001; Cook & Brown, 1999; Lave & Wenger, 1991; Orlikowski, 2002, 2006; Spender, 1994). This alternative *practice-based perspective* asserts that affixing knowledge (as an object) to individuals or groups obscures what it means *to know*. In practice-based thinking, "knowing" is a reference to competent participation in action occurring in organizational scenes populated by heterogeneous arrays of discursive and material elements (Kuhn & Porter, 2011; Østerlund & Carlile, 2005). Knowledge, in turn, cannot be isolated or transferred, because identities, sites, histories, artifacts, ideas, technologies, bodies, discourses, and the like are all intimately bound together in organizing practice.

Knowledge, as understood in the literatures we have reviewed, is not erased in a practice-based perspective; instead, it becomes "*a tool at the service of knowing*, not something that, once possessed, is all that is needed to enable action or practice" (Cook & Brown, 1999: 388. emphasis in original). The practice and the product are thus intertwined. Practice-based perspectives, then, tend to be concerned that conventional accounts of KM amount to little more than "add knowledge and stir" thinking: Emphasizing knowledge as a commodity, as an object, leads scholars to believe in the importance of sharing and locating it, but neglects an understanding of how it becomes understood as valuable, how its benefits are achieved, and how power operates subtly in processes of knowing (Contu & Willmott, 2003; Marabelli & Newell, 2014).

A practice-based perspective on knowing, as an alternative to the dominant object-based conception of knowledge, makes several claims on the KM literature. These claims hinge on a shift toward the centrality of *how* questions in KM, as contrasted with *which, what,* and *where* questions. As demonstrated previously, the overriding concerns in KM (at least in the extant literature) are with identifying which knowledge is to be shared, which variables are associated with organizational performance, what forms integration and differentiation should take, and where valuable knowledge is located. The practice-based view, by contrast, directs attention to how myriad elements are configured to produce knowledgeable action, how learning is evident in altered practices, how expertise emerges (and how it is made to matter), and how contradictions in a practice (or a constellation of practices) are managed. *How* questions, in other words, lead analyses in directions that differ from conventional KM literature. Yet the question confronting this practice-based vision is not merely whether it asks different questions, but whether it can produce novel conceptual insights and guidance for practice that matter to KM scholars, especially those working from a strategic management frame. In the remainder of this section, we sketch two specific guides for KM research based on a practice perspective, with an eye toward developing relevant novelty.

"Invisible" knowledge work. A first research direction suggested by a practice-based perspective is the illumination of previously hidden labor in the conduct of knowledge work. Interpretive studies of organizations have shown that, when we conceptually reduce organizing to interactions between a select set of variables, our analytical gaze obscures a good deal of work without which routine action could not proceed. Although not always conducted under the aegis of KM research, practice-based studies display how invisible, or hidden, work makes possible the work that is observed. Invisible work includes the subtle moves that evince unstated rules (Llewellyn & Hindmarsh, 2013; Suchman, 2000), private and unpaid (and thus often gendered) labor (Daniels, 2014; Smith, 2001), "articulation work" that adjusts and monitors interaction to get activity "back on track" after unexpected events (Strauss, Fagerhaugh, Suczek, & Wiener, 1985), "process" work that is not easily aligned with the prescribed work of professions (Treem & Barley, 2016), and work which is in some sense distant from a focal activity (Leonardi, Huysman, & Steinfeld, 2013). A practice perspective encourages attention to this invisible labor, including the work performed by nonhuman participants such as KM technologies. One route to pursue invisible labor in KM would be to examine efforts to shape the metrics and indicators by which work is evaluated. As critical accounting scholars have noted, control over these metrics and indicators is key to defining what counts as work in the first place and are sites in which struggles over meaning can be observed (Power, 2001). Particularly in the case of knowledge work, which is less directly observable and assessable than other forms of labor, shaping the recognition of when work occurs is directly relevant not merely to guiding the work of innovation, but also to monitoring, controlling, and rewarding (or punishing) workers (Star & Strauss, 1999). Thus, potential research questions for this sort of direction would ask (a) *how the invisible interactive work conducted by human and nonhuman participants develops and promulgates assessments (i.e., metrics and indicators) of knowledge and knowledge work*; (b) *how, through those interactions, particular interests become inscribed into those assessments*; and (c) *whether particular forms of practice are associated with an emphasis on either common ground or specialization*.

Heterogeneity in knowledge and knowing. A second practice-based research direction for KM scholarship is to examine the importance of differentiation, via the exposure to difference, in organizational practice. For instance, Bruns's (2013) aforementioned study of a dispersed systems biology research program showed collaboration and coordination to be highly complex and contextualized processes. They occurred both within and across domains only when situated practices—both individual and collective—were in place. In contrast to the assertion in much of the KM literature that *integrated* knowledge is required to generate the collaboration and coordination associated with organizational innovation, Bruns' study suggests that exposure to work in other areas, as a display of *differentiation*, can be an important contributor to coordination. She calls, then, for tactics that can aid actors in appreciating differences, including shadowing, job rotation, workshops, and training in communicative practices that foster cross-domain dialogue; these tactics retain each domain's unique knowledge and foster an understanding of the differences that enable a division of labor to operate successfully. Foregrounding forms of communicative practice in the pursuit of coordination and collaboration can aid in overcoming the conception of knowledge as a commodity that can travel easily from place to place and, once received in the desired location, will produce frictionless action. Practice-based thinking highlights the ever-present potential for schisms and breakdowns, along with the array of sub-practices required to accomplish the processes often taken for granted in the KM literature. Research in this vein would investigate how heterogeneity in knowledge and knowing—in the sense of both multiple individuals and multiple trajectories of practice in a single site (Schatzki, 1996)—has the potential to generate not merely conflict and ambiguity (as a conventional conception of KM might suggest), but also new conceptual frames for the practice in question (Kuhn & Porter, 2011). A research aim pursuing the generativity of tensions and paradoxes resulting from the heterogeneity of knowledge and knowing must be accompanied by an understanding that reframing is not always instrumentally beneficial or managerially desirable. Indeed, reframing often is mediated by, and is the outcome of, significant intraorganizational conflicts (Dewulf et al., 2009; Fairhurst, 2007; Kuhn & Corman, 2003) Reframing may thus lead to outcomes unaligned with strategic aims and may just as easily close off as open up efforts to innovate. In short, heterogeneity and reframing must be *managed*, but efforts to produce order out of heterogeneity are likely to carry unintended consequences (Vásquez, Schoeneborn, & Sergi, 2016).

Beyond this, a related line of thinking acknowledges that, for organizations operating in complex and turbulent environments, organizational *ambidexterity* may be beneficial. Scholarship considering the differential absorptive capacity of organizations and groups recognizes the importance of a firm's ability to integrate, assimilate, and act upon internal and external sources of knowledge (Lane et al., 2001; Marabelli & Newell, 2014; Volberda et al., 2010). In this light, ambidexterity refers to the capacity to absorb knowledge through a balance of exploration and exploitation, enabling business units to both enact adaptability and alignment with organizational goals

(Gibson & Birkinshaw, 2004; Raisch, Birkinshaw, Probst, & Tushman, 2009). Although work on ambidexterity has been tremendously helpful for insights into how organizations pursue multiple goals, either in cyclical or simultaneous fashion, it provides less insight on cases in which stakeholders, both internal and external, may use conflicting evaluative criteria—which are common elements of the contemporary organizational landscape (Hsu & Hannan, 2005)—in evaluating organizational action. What complex organizations require in such conditions, suggests Stark (2009), are practices that foster a capacity to speak to multiple evaluative principles simultaneously: The organizational imperative is to "*keep multiple evaluative principles in play and to exploit the resulting friction of their interplay*" (p. 15, emphasis in original). From this perspective, KM initiatives that emphasize integration (i.e., common ground) run the risk of being able to appeal to only a limited range of evaluative principles; this restriction, in turn, works at cross-purposes with their desire to manage exogenous and endogenous uncertainty. The notion of exploiting the friction of evaluative principles' interplay suggests, instead, a need for organizational practices that can (a) display the value of organizational action for manifold stakeholder groups (Gehman et al., 2013) and (b) reframe conventional practices to embrace alternative modes of knowing as a route to recognizing possibilities for innovation (Nicholls, 2009). Research on this theme might, then, inquire about *how knowledge heterogeneity produces practices that reframe and/or challenge existing KM initiatives, with an eye toward how efforts to engage heterogeneity foster or inhibit the innovative management of integration–differentiation tensions*.

Knowledge Management as a Form of Ongoing Organizational Change

A strength of seeing KM through a practice lens is the recognition of the dynamic character of knowledge. One reason why the integration bias described earlier limits our understanding of organizational knowledge and KM is that it risks presenting knowledge as a static commodity, a resource useful at a particular time. Even in situations where the espoused goal is knowledge creation and integration, the benefits of producing common ground are motivated by the idea is that knowledge, once made manifest, is of a distinct form that is useful for the organization. As a result, knowledge trajectories in organizations have largely been presented as unidimensional (treating knowledge in a singular form) and unidirectional (knowledge moves in a particular direction). This approach is attractive because it lends itself to variance approaches (Poole & Van de Ven, 1989) that treat knowledge as a stable and predictable input to, or output of, an organizational system. KM initiatives can then be conceptualized as failures or successes based on clear assessment criteria (Schultze & Leidner, 2002). However, variance approaches to KM presents two limitations: (a) they discount the dynamic nature of knowledge and the myriad ways that knowledge is managed over time, as presented in the preceding subsections; and (b) they obscure the possibility of differential knowledge trajectories, and differential outcomes, at different levels of analysis.

Yet, to say that knowledge is a dynamic construct will not offer analytic utility unless we take efforts to understand *how* changes in knowledge take place in organizational contexts over time. Applying concepts of organizational change, while viewing KM activities themselves as the objects of analysis, may offer a vantage point to pursue understanding of such dynamism (Van de Ven & Poole, 1995). Commonly, KM research offers individuals, groups, or some operationalized representation of knowledge as the unit of analysis, applying survey methods that use cross-sectional data to determine relationships between the level (or type) of knowledge and some feature of collective performance (Cummings, 2004; Gold, Malhotra, & Segars, 2001; Lee & Choi, 2003; Levin & Cross, 2004; Szulanski, 2000). This approach perpetuates an integration bias in part because it requires the researcher to declare the knowledge trajectory of interest before analysis (e.g., differentiation to integration), a conceit that often favors the study of integrative processes. Alternately, studying KM initiatives and related activities across time and across levels of analysis does not require an a priori assumption regarding any particular knowledge trajectory and allows the possibility of exploring multiple knowledge trajectories simultaneously. Recognizing the inherent dynamism of knowledge in organizations, scholars have acknowledged that even organizational routines, which are viewed as a means of institutionalizing organizational knowledge and facilitating predictable actions, are sources of continuous change (Feldman & Pentland, 2003). As a result, an organization's capabilities to manage knowledge, as well as its KM needs, will shift based on both structural and interactional alterations in the organization over time (Argote, McEvily, & Reagans, 2003).

A change-oriented perspective on KM is consistent with a perspective that views organizations as constituted by systems of knowledge (Holzner & Marx, 1979). As Pentland (1995: 5) noted regarding processes of knowledge, "The idea is that knowledge is the product of an ongoing set of practices embedded in the social and physical structures of the organization. It is meant to convey the dynamic quality of the overall system." A systems approach recognizes KM as the ongoing interplay of actors, artifacts, and activity that creates an interdependent network of relations (Blackler, 1995; Garud & Kumaraswamy, 2005; Holzner & Marx, 1979; Pentland, 1995; Spender, 1996). This perspective is attractive because it recognizes the dynamic ways knowledge is used, recognizes a role for artifacts and nonhuman actors in processes of knowledge, and looks at KM as occurring both within specific relations (components of the system) and a larger organizational network. In discussing ways to study processes of change in organizations, Langley, Smallman, Tsoukas, and Van de Ven (2013) note the importance of considering two elements in an analysis: Time and changes in activity. Integrating time and temporality into the study of KM will inform how changes in the organizational environment influence KM processes, help examine KM initiatives from development through implementation and use, and consider changes in the nature of KM technologies over time. In addition, looking at KM systems as networks of action and activity will help reveal KM processes at multiple levels of organizations, incorporate nonhuman actors into the study of KM, and recognize how the visible articulation of network relations can alter KM processes.

Temporality. Change takes place over time, and examining KM as a longitudinal process will help capture the form and frequency of alterations in knowledge trajectories. Recognizing the mechanisms of change present in KM, scholars have represented KM in organizations in terms of stages (Szulanski, 2000) or cycles (Birkinshaw & Sheehan, 2002; Garud & Kumaraswamy, 2005). These approaches recognize that organizational uses for knowledge in an organization may shift and knowledge processes may develop an inertia, or path dependency, that affects their predictability and susceptibility to change. Furthermore, organizations may, over time, develop resources or build on experiences in ways that influence KM outcomes (Gold et al., 2001). Acknowledging this dynamism offers an analytical space in which trajectories of integration or differentiation may emerge.

Attewell's (1992) description of how business computing knowledge diffused through organizations over decades demonstrates the value of incorporating temporality into the study of how organizations acquire, learn, and integrate knowledge. Early on, when business computing was both novel and resource intensive, it was strategically advantageous to outsource computing tasks to specialist service organizations. Over time, firms garnered computer gurus and developed departments with specialized knowledge in business computing. As knowledge of business computing became more common, these specialist departments became marginalized and started to dissolve amidst growing expectations that workers possess a knowledge base adequate for self-service. Analyzing changes in how organizations managed knowledge over decades revealed a shift from actively sustaining differentiated knowledge to working to integrate knowledge; doing so also displayed how changes in trajectories were intertwined with shifts in available resources, including knowledge gained over time.

Another benefit of implementing temporality into the study of KM and organizational change is that it can foreground the role and intentions of management in KM processes. Although KM is a concept frequently invoked in the study of organizational knowledge, there is a disproportionate focus on knowledge outcomes and little explicit recognition of actions taken by organizations to manage that knowledge (Alvesson & Kärreman, 2001). A change-oriented perspective means that KM can be analyzed both in relation to distinct purposeful decisions of organizational authorities and in terms of ongoing emergent influences that alter organizational knowledge. Taking time into consideration encourages analysts to cross the "implementation line" (Leonardi, 2009) and ask questions related to how the development of KM initiatives may contribute to subsequent organizational changes. As an example, Garud and Kumaraswamy (2005) traced KM initiatives at a software services company where management sought to capture and codify the experiences of workers, eventually leading to the development of a central KM portal that housed workers' contributions. Seeking to spur participation with the portal management created an incentive for contributions to the KM system, which resulted in a spike in content added, but a low quality in knowledge provided. Subsequently, individuals searching for information became deterred by the need to sort through the wealth of the material in the KM system and decreased their use of the portal, creating a "vicious cycle" of (dis)use. It was only in understanding the decisions of management in implementing the incentive and in investigating the unfolding of the initiative over time that this study was able to ascertain this unintended (and interesting) consequence.

Garud and Kumaraswamy's study also highlights the role that time can play in altering local meanings for, and uses of, KM technologies. In KM initiatives, technologies such as portals can be viewed as public goods in that they offer a resource available to all, but one to which not all individuals are obligated to contribute (Fulk, Flanagin, Kalman, Monge, & Ryan, 1996). As such, KM technologies are perpetually at risk of failure due to lack of contributions, maintenance, and use. KM scholarship has examined the dilemma of eliciting individual contributions to communal knowledge repositories and found that even when KM initiatives overcome obstacles to initial contributions, they face problems associated with crowdedness and perceived value as they extend in time (Kankanhalli et al., 2005; Ma & Agarwal, 2007; Wasko & Faraj, 2005). As contributions to repositories build, it can be increasingly difficult to find specific information (Merritt, Ackerman, & Hung, 2016). More broadly, the meanings (and sets of possibilities) of the KM system change over time as it is used by organizational members. Only by viewing KM initiatives longitudinally can we see the ways in which the system evolves in terms of use and usefulness, and identify when technologies designed for integrating knowledge may be understood as (in)capable of meeting that goal—or if they elicit alternative actions that support differentiated knowledge. Particularly, when KM initiatives and technologies are used across multiple organizational units and levels, differing interpretations and uses can create challenges to coordination; if managerial expectations revolve around integrated knowledge and seamless coordination across those units or levels, shifts in use over time can create operational challenges. Developing a better understanding the intentions of management in designing and implementing KM systems, along with the sociomaterial ways these systems change over time (irrespective of managerial intent), can provide a richer understanding of organizational appropriations of KM systems. This line of inquiry, then, would advance questions about *whether and why interpretations and uses of KM tools shift over time, and how those changes generate challenges to efforts to create knowledge integration or differentiation.*

Knowledge networks. An additional route to (re) conceptualizing KM is to start with the *knowledge network*, a vision of the distribution of knowledge across interconnected participants. The notion of a knowledge network moves away from conceptualization of organizational knowledge as an aggregate stock of knowledge located in individuals and routines, and embraces the notion that collective knowledge exists in the links between knowledge elements relevant to organizing (Contractor & Monge, 2002; Hansen, 2002; Jarvenpaa & Majchrzak, 2008; Wasko & Faraj, 2005). Phelps et al. (2012) portray these knowledge networks as a set of nodes, "individuals or higher level collectives that serve as heterogeneously distributed repositories of knowledge and agents that search for, transmit, and create knowledge—interconnected by social relationships that enable and constrain nodes' efforts to acquire, transfer, and create knowledge" (p. 1117). Yet these nodes need not be limited to human actors and bits of information; as Contractor, Monge, and Leonardi (2011) note knowledge networks can be both *multimodal*, meaning that there is likely to be a wide variety of participants in the network (e.g., persons, artifacts, documents, technologies, etc.), and *multiplex*, meaning that multiple types of relationships among nodes can be represented. Put differently, an object such as a knowledge repository, an expert recommender system, or an online document can be conceptualized as providing knowledge to or receiving knowledge from organizational activity.

If knowledge networks are multimodal and multiplex, they also span levels of analysis. Although conceptually attractive, tracking knowledge networks across levels of analysis is analytically challenging (Hitt, Beamish, Jackson, & Mathieu, 2007; Morgeson & Hofman, 1999). Yet investigating cross-level relationships is important because the composite of micro-level processes exhibits emergent outcomes that can create group-level effects; at the same time, collective structures constrain and enable lower level processes (Kozlowski & Klein, 2000). Within the contexts of KM, this cross-level relationship has been useful for analyzing the allocation and retrieval of knowledge among organizational experts in group contexts (Yuan et al., 2010), where individuals' decisions to share or exchange knowledge are not only influenced both by their individual attributes and network position, but also attributes and structures at the level of the network. Novel statistical analyses such as exponential random graph modeling (Contractor, Wasserman, & Faust, 2006) and agent-based modeling (Palazzolo, 2005) allow for simulations or the use of large-scale data sets that test multiple theoretical mechanisms simultaneously across multiple levels of network data. Capturing multiple levels of network influences provides a fuller picture of the drivers of change in KM systems.

In addition, framing KM in network terms directs attention to the roles meta-knowledge may play in organizational processes. Specifically, KM initiatives (and, in particular, the use of KM technologies that provide directories of relationships or expertise) can develop understandings of "who knows what" and "who knows who" in organizations (Jackson & Klobas, 2008; Nevo & Wand, 2005). This meta-knowledge can make the work and expertise of others more visible and articulate organizational networks that were previously obscured (Cross, Borgatti, & Parker, 2002). The importance of meta-knowledge is based on a recognition that KM creates more than a static network of organizational knowledge, and can facilitate a context for active *networking* in organizations in which individuals seek to influence perceptions of knowledge held by coworkers, and to locate new sources of insight in the pursuit of innovations (DiMicco, Geyer, Millen, Dugan, & Brownholtz, 2009). Scholars have recognized that studying the consequences of this network articulation is particularly important given the increased adoption of social media technologies within organizational contexts (Kane, Alavi, Labianca, & Borgatti, 2014; Treem & Leonardi, 2012). As KM technologies, such as social media, make associations among individuals or content visible, they may alter the ways workers chose to communicate knowledge with others (Leonardi & Treem, 2012), or create a competitive marketplace for knowledge resources (Hansen & Haas, 2001). Analyses guided by a focus on knowledge networks might then ask *how different actors and relations engaged in KM might play distinct roles in processes of knowledge integration and differentiation, how KM processes at one level of a knowledge network might influence process at another level, and how meta-knowledge of a knowledge network facilitated by KM influences processes of knowledge integration and differentiation*.

CONCLUSION

In our introduction, we observed that knowledge-based theories of the firm are closely coupled, but distinct from the field of KM. Although KM research has largely evolved from the shift toward knowledge-based theory, we have shown that KM scholarship over the past two decades has privileged particular trajectories of knowledge over others. Although we have compelling evidence that integrational processes play an important role in measurable organizational outcomes, we believe an important step forward for KM research will involve broadening our analytic scope to include a more dynamic vantage of managing knowledge in organizational contexts. Incorporating a multitude of knowledge trajectories into the study of KM, and taking seriously the potential value of differentiation, offers new opportunities for scholarship to explore the ways in which KM can provide organizational value.

As a final comment, we would like to reflect on the nature of knowledge in today's organizational contexts as it relates to the context of organizing when the field of KM evolved in the 1990s. As we have evolved firmly into an information and knowledge economy, the primary problems of knowledge and organizing may be in the midst of important shifts. A growing number of social and technical changes have afforded organizations an increasing capacity for leveraging information, knowledge resources, and domain experts such that *knowledge access* may no longer be the central problem that organizations face. The presence of massive data repositories, records of consumer behavior, and the ability to establish and maintain ties with individuals through online social networks means that workers have greater abilities to access knowledge resources or interact with knowledgeable others. Finding opportunities to aggregate or access knowledge is much less the problem than is managing the fire hose. The problems in a context of high knowledge access are meaningfully distinct: the important questions shift from gathering as much knowledge as possible to figuring out when to *stop* gathering information. They shift from getting everything you can to *evaluating which* knowledge is *worthy* of inclusion. At its heart, the current era seems to involve increased cachet for knowledge trajectories beyond integrational processes. However, limiting opportunities for information access and knowledge development may be difficult; once the fire hose is turned on, it may be hard to convince people not to keep drinking. Individuals may continue to seek available knowledge beyond the point it is beneficial for decision-making for symbolic reasons, so that others in the organization will view them as acting in a responsible manner (Feldman & March, 1981). Although advances in technologies and investments in

KM appear to favor opportunities for integrative trajectories, it remains to be seen how organizations might effectively develop technological, structural, or relational strategies to help restrict or filter knowledge access in some manner.

This trend exemplifies the point that knowledge differentiation should be recognized as much more than the mere presence of unintegrated knowledge. Rather knowledge differentiation is a trajectory of knowledge to which organizations should consider focusing KM strategy and associated practices. Just as knowledge integration involves active work to overcome the obstacles associated with the presence of diverse sets of knowledge in organizations, so too does knowledge differentiation require purposeful effort and structural support to provide space to establish and maintain specialized work. Greater focus on knowledge differentiation broadens the potential value associated with KM in organizations, embraces calls for dynamism in the study of organizational knowledge, and promotes a research agenda for KM that captures the breadth of knowledge trajectories present in contemporary work.

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APPENDIX: LIST OF STUDIES INCLUDED IN THE STRUCTURED LITERATURE REVIEW

The studies below are listed chronologically by year and ranking within that year. “Total Results” reflects the total number of responses returned from our search string for that particular year. “Times cited” indicates the number of sources citing the study at the time of data collection. The presence of an asterisk in the final four columns indicates which of the four knowledge trajectories we coded as being present within each article. A few articles do not have trajectories assigned to them because they did not fit into our typology. We include those studies here for completeness.

					Knowledge Trajectories Represented			
Year	Total Results	Rank	Times Cited	Citation	Dif ↓ Int	Dif ↓ Dif	Int ↓ Dif	Int ↓ Int
1996	86	1	2338	Szulanski (1996)	*			
1996	86	2	1627	Grant (1996a)	*			
1996	86	3	1176	Spender (1996)	*	*	*	*
1996	86	4	1030	Mowery, Oxley, and Silverman (1996)	*			
1996	86	5	973	Kogut and Zander (1996)	*	*	*	*
1996	86	6	714	Sanchez and Mahoney (1996)	*	*		
1996	86	7	443	Henderson and Cockburn (1996)	*			
1996	86	8	398	Star and Ruhleder (1996)	*	*		
1996	86	9	373	Liebeskind, Oliver, Zucker, and Brewer (1996)	*			
1996	86	10	338	Spender and Grant (1996)				
1997	100	1	773	Hargadon and Sutton (1997)	*			
1997	100	2	226	Achrol (1997)	*	*		
1997	100	3	198	Lam (1997)	*			
1997	100	4	180	Warkentin, Sayeed, and Hightower (1997)	*			
1997	100	5	158	Sharma (1997)	*	*		
1997	100	6	158	Wiig (1997)	*		*	
1997	100	7	153	Demarest (1997)	*		*	
1997	100	8	121	Wiig (1997)	*		*	
1997	100	9	119	GreganPaxton and John (1997)	*			
1997	100	10	118	Quintas, Lefrere, and Jones (1997)	*			
1998	131	1	3020	Dyer and Singh (1998)	*			
1998	131	2	718	Davenport, De Long, and Beers (1998)	*			
1998	131	3	389	O’Dell and Grayson (1998)	*			
1998	131	4	364	Poppo and Zenger (1998)	*			
1998	131	5	341	Zucker, Darby, and Armstrong (1998)	*			
1998	131	6	320	Madhavan and Grover (1998)	*			
1998	131	7	315	Ruggles (1998)	*			
1998	131	8	310	Inkpen and Dinur (1998)	*			
1998	131	9	210	DeFillippi and Arthur (1998)	*			
1998	131	10	197	Fahey and Prusak (1998)	*			
1999	165	1	966	Hansen, Nohria, and Tierney (1999)	*			
1999	165	2	643	Simonin (1999)	*			
1999	165	3	508	Zack (1999)	*			
1999	165	4	431	DeCarolis and Deeds (1999)	*		*	
1999	165	5	312	Bresman, Birkinshaw, and Nobel (1999)	*			
1999	165	6	283	McDermott (1999)	*			
1999	165	7	212	Malone et al. (1999)	*		*	
1999	165	8	193	Simonin (1999)	*			
1999	165	9	181	Bontis (1999)	*		*	
1999	165	10	130	Sarvary (1999)	*		*	
2000	228	1	1069	Gupta and Govindarajan (2000)	*			
2000	228	2	1032	Dyer and Nobeoka (2000)	*			
2000	228	3	884	Kale, Singh, and Perlmutter (2000)	*			
2000	228	4	783	Argote and Ingram (2000)	*			
2000	228	5	633	Wenger and Snyder (2000)	*			

APPENDIX
(Continued)

					Knowledge Trajectories Represented			
Year	Total Results	Rank	Times Cited	Citation	Dif ↓ Int	Dif ↓ Dif	Int ↓ Dif	Int ↓ Int
2000	228	6	501	Osterloh and Frey (2000)	*	*	*	
2000	228	7	481	Szulanski (2000)	*	*		
2000	228	8	474	Wasko and Faraj (2000)	*			
2000	228	9	387	De Long and Fahey (2000)	*		*	
2000	228	10	354	Rivkin (2000)	*	*		
2001	298	1	2031	Alavi and Leidner (2001)	*		*	
2001	298	2	1004	Tsai (2001)	*			
2001	298	3	712	Gold, Malhotra, and Segars (2001)	*		*	*
2001	298	4	581	Lane, Salk, and Lyles(2001)	*			
2001	298	5	567	Cramton (2001)	*	*		
2001	298	6	489	Lu and Beamish (2001)	*			
2001	298	7	419	Brusoni, Prencipe, and Pavitt (2001)	*	*	*	
2001	298	8	324	Winter and Szulanski (2001)	*		*	
2001	298	9	305	Earl (2001)	*			
2001	298	10	291	Tsoukas and Vladimirou (2001)	*	*	*	
2002	280	1	803	Carlile (2002)	*	*		
2002	280	2	506	Calantone, Cavusgil, and Zhao (2002)	*			
2002	280	3	503	Tsai (2002)	*			
2002	280	4	462	Hansen (2002)	*	*		
2002	280	5	357	Ireland, Hitt, and Vaidyanath (2002)	*			
2002	280	6	356	Zollo, Reuer, and Singh (2002)	*			
2002	280	7	286	Cabrera and Cabrera (2002)	*			
2002	280	8	259	Markus, Majchrzak, and Gasser (2002)	*			
2002	280	9	253	Bhagat, Kedia, Harveston, and Triandis (2002)	*			
2002	280	10	245	Okhuysen and Eisenhardt (2002)	*			
2003	344	1	890	Reagans and McEvily (2003)	*		*	
2003	344	2	589	Argote, McEvily, and Reagans (2003)	*			
2003	344	3	555	Gertler (2003)	*	*		
2003	344	4	442	Bechky (2003)	*			
2003	344	5	432	Lee and Choi (2003)	*			
2003	344	6	414	Rosenkopf and Almeida (2003)	*			
2003	344	7	322	Uzzi and Lancaster (2003)	*			
2003	344	8	317	Kotabe, Martin, and Domoto (2003)	*			
2003	344	9	260	Song, Almeida, and Wu (2003)	*			
2003	344	10	242	Gibson and Vermeulen (2003)	*	*	*	
2004	356	1	712	Levin and Cross (2004)	*			
2004	356	2	423	Cummings (2004)	*			
2004	356	3	318	Nickerson and Zenger (2004)	*	*		
2004	356	4	300	Pittaway, Robertson, Munir, Denyer, and Neely (2004)	*			
2004	356	5	289	Hatch and Dyer (2004)	*	*		
2004	356	6	273	Oxley and Sampson (2004)	*	*		
2004	356	7	272	Tallman, Jenkins, Henry, and Pinch (2004)	*	*		
2004	356	8	270	McFadyen and Cannella (2004)	*			*
2004	356	9	264	Dhanaraj, Lyles, Steensma, and Tihanyi (2004)	*			
2004	356	10	262	Beckman, Haunschild, and Phillips (2004)	*			*
2005	458	1	1028	Wasko and Faraj (2005)	*			
2005	458	2	890	Bock, Zmud, Kim, and Lee (2005)	*	*		
2005	458	3	821	Inkpen and Tsang (2005)	*			
2005	458	4	643	Kankanhalli, Tan, and Wei (2005)	*			
2005	458	5	320	Smith, Collins, and Clark (2005)	*			
2005	458	6	320	Ko, Kirsch, and King (2005)	*			

APPENDIX
(Continued)

Year	Total Results	Rank	Times Cited	Citation	Knowledge Trajectories Represented			
					Dif ↓ Int	Dif ↓ Dif	Int ↓ Dif	Int ↓ Int
2005	458	7	273	Singh (2005)			*	
2005	458	8	251	Levina and Vaast (2005)	*			
2005	458	9	247	Malhotra, Gosain, and El Sawy (2005)	*		*	*
2005	458	10	229	Hansen, Mors, and Løvås (2005)	*			
2006	546	1	603	Chiu, Hsu, and Wang (2006)	*			
2006	546	2	523	Lane, Koka, and Pathak (2006)				
2006	546	3	424	Collins and Smith (2006)	*			
2006	546	4	321	Lavie and Rosenkopf (2006)	*			*
2006	546	5	321	McAfee (2006)	*			*
2006	546	6	316	Dhanaraj and Parkhe (2006)	*	*		
2006	546	7	276	Dyer and Hatch (2006)	*			
2006	546	8	273	Krishnan, Martin, and Noorderhaven (2006)				
2006	546	9	247	Srivastava, Bartol, and Locke (2006)	*			*
2006	546	10	237	Pavlou and El Sawy (2006)				
2007	671	1	288	Jensen, Johnson, Lorenz, and Lundvall (2007)	*			
2007	671	2	286	Sampson (2007)	*	*		
2007	671	3	278	Wang and Ahmed (2007)	*		*	
2007	671	4	274	Mitton, Adair, McKenzie, Patten, and Perry (2007)				
2007	671	5	254	D'Este and Patel (2007)		*	*	
2007	671	6	222	De Luca and Atuahene-Gima (2007)	*			
2007	671	7	221	Perkmann and Walsh (2007)	*			
2007	671	8	219	Ma and Agarwal (2007)	*	*		
2007	671	9	216	Lin (2007)	*			
2007	671	10	213	Fleming, Mingo, and Chen (2007)	*			*
2008	899	1	212	Chow and Chan (2008)	*			
2008	899	2	208	van Wijk, Jansen, and Lyles (2008)	*			
2008	899	3	193	Amin and Roberts (2008)	*			*
2008	899	4	162	Easterby-Smith, Lyles, and Tsang (2008)	*			
2008	899	5	135	Dibbern, Winkler, and Heinzl (2008)	*			
2008	899	6	129	Tiwana (2008)	*			*
2008	899	7	122	Bekkers and Freitas (2008)		*		
2008	899	8	108	Im and Rai (2008)	*			*
2008	899	9	108	Singh (2008)	*			
2008	899	10	108	Monteiro, Arvidsson, and Birkinshaw (2008)	*			
2009	1061	1	245	Christian, Bradley, Wallace, and Burke (2009)				*
2009	1061	2	207	Lichtenthaler (2009)	*			
2009	1061	3	197	Mesmer-Magnus and DeChurch (2009)	*			
2009	1061	4	192	Breschi and Lissoni (2009)	*			
2009	1061	5	174	Liao, Toya, Lepak, and Hong (2009)		*		
2009	1061	6	171	Chen and Huang (2009)				*
2009	1061	7	160	Lichtenthaler and Lichtenthaler (2009)	*	*		
2009	1061	8	144	Escribano, Fosfuri, A., and Tribó (2009)	*			
2009	1061	9	137	Kuenzi and Schminke (2009)	*			*
2009	1061	10	133	Tsoukas (2009)			*	
2010	1195	1	263	Wang and Noe (2010)	*			
2010	1195	2	192	Volberda, Foss, and Lyles (2010)	*			
2010	1195	3	188	Raymond, Fazey, Reed, Stringer, Robinson, and Evely (2010)	*			
2010	1195	4	171	Lavie, Stettner, and Tushman (2010)	*	*		
2010	1195	5	160	Phelps (2010)		*		
2010	1195	6	149	Lee, Park, Yoon, and Park (2010)		*		

APPENDIX
(Continued)

Year	Total Results	Rank	Times Cited	Citation	Knowledge Trajectories Represented			
					Dif ↓ Int	Dif ↓ Dif	Int ↓ Dif	Int ↓ Int
2010	1195	7	136	Bruneel, D'Este, and Salter (2010)		*		
2010	1195	8	106	Zheng et al. (2010)	*			
2010	1195	9	100	Contandriopoulos, Lemire, Denis, J.-L., and Tremblay (2010)		*		
2010	1195	10	91	Sosna, Trevinyo-Rodríguez, and Velamuri (2010)	*			
2011	1284	1	194	Nahrgang, Morgeson, and Hofmann (2011)		*		
2011	1284	2	184	Meyer, Mudambi, and Narula (2011)	*			
2011	1284	3	156	Argote and Miron-Spektor (2011)	*			
2011	1284	4	156	Leonardi (2011)		*		
2011	1284	5	127	Chandler and Vargo (2011)	*			
2011	1284	6	126	Jiménez-Jiménez and Sanz-Valle (2011)	*			
2011	1284	7	120	Yates and Paquette (2011)	*			
2011	1284	8	107	Xu (2011)	*			
2011	1284	9	102	Ordanini and Parasuraman (2010)	*			
2011	1284	10	102	Chang and Chuang (2011)	*			
2012	1343	1	81	Felin, Foss, Heimeriks, and Madsen (2012)	*			
2012	1343	2	73	Zhou and Li (2012)	*			
2012	1343	3	68	Chang, Gong, and Peng (2012)	*			
2012	1343	4	66	Kumaraswamy, Mudambi, Saranga, and Tripathy (2012)		*		
2012	1343	5	63	Luo, Huang, and Wang (2012)	*			
2012	1343	6	60	Broekel and Boschma (2012)	*			
2012	1343	7	60	von Krogh (2012)		*		
2012	1343	8	60	Cheung an Lee (2012)	*			
2012	1343	9	58	Susanne and Ingi Runar (2012)	*			
2012	1343	10	52	García-Morales, Jiménez-Barrionuevo, and Gutiérrez-Gutiérrez (2012)	*			
2013	1431	1	104	Chandrasegaran et al. (2013)		*		
2013	1431	2	49	Majchrzak, Faraj, Kane, and Azad (2013)			*	
2013	1431	3	47	Petter, deLone, and McLean (2013)		*		
2013	1431	4	43	Lorenzen and Mudambi (2013)	*			
2013	1431	5	41	Mergel (2013)		*		
2013	1431	6	39	Ritala and Hurmelinna-Laukkanen (2013)	*	*		
2013	1431	7	35	Majchrzak, Faraj, and Azad (2013)	*			
2013	1431	8	34	Davison, Ou, and Martinsons (2013)	*			
2013	1431	9	32	Fazey et al. (2012)	*			
2013	1431	10	32	Amayah (2013)	*			
2014	1477	1	45	Kane, Alavi, Labianca, and Borgatti (2014)	*			
2014	1477	2	36	Beske, Land, and Seuring (2014)	*			
2014	1477	3	27	Reed, Stringer, Fazey, Evely, and Kruijsen (2014)	*			
2014	1477	4	27	Balliet, Wu, and De Dreu (2014)	*			
2014	1477	5	27	Ganter and Hecker (2014)		*		
2014	1477	6	25	Schrettle, Hinz, Scherrer -Rathje, and Friedli (2014)		*		
2014	1477	7	22	Phillips, Pullen, and Rhodes (2013)				
2014	1477	8	20	Ma and Chan (2014)	*			
2014	1477	9	20	Leonardi (2014)	*			
2014	1477	10	18	Brannen, Piekkari, and Tietze (2014)	*			
2015	1561	1	25	Lusch and Nambisan (2015)	*			
2015	1561	2	16	Navimipour, Rahmani, Habibizad Navin, and Hosseinzadeh (2015)		*		
2015	1561	3	16	Ritala, Olander, Michailova, and Husted (2015)	*			

APPENDIX
(Continued)

Year	Total Results	Rank	Times Cited	Citation	Knowledge Trajectories Represented: Dif → Int	Dif → Dif	Int → Dif	Int → Int
2015	1561	4	15	Palacios-Marqués, Soto-Acosta, and Merigó (2015)		*		
2015	1561	5	14	Barrett, Davidson, Prabhu, and Vargo (2015)	*			
2015	1561	6	12	Zhang, Gao, Yan, de Pablos, Sun, and Cao (2015)	*			
2015	1561	7	12	Donate and Sánchez de Pablo (2015)		*		
2015	1561	8	11	Ngai, Moon, Lam, Chin, and Tao (2015)	*			
2015	1561	9	10	Kimmerle, Moskaliuk, Oeberst, and Cress (2015)			*	
2015	1561	10	10	Duffield and Whitty (2015)	*			